

RAG (Retrieval-Augmented Generation) - Fundamentals

Retrieval-Augmented Generation (RAG) is a machine learning technique that combines information retrieval with generative AI models to produce more accurate, contextual, and grounded responses. RAG addresses a fundamental limitation of large language models: their knowledge cutoff dates and tendency to generate plausible-sounding but factually incorrect information, a phenomenon known as "hallucination." By augmenting generative models with the ability to retrieve relevant information from external knowledge sources, RAG creates a hybrid system that leverages the strengths of both retrieval and generation. The technique was first introduced by Facebook AI (Meta) researchers in 2020 and has since become a foundational approach in modern NLP applications. RAG systems are particularly valuable for domain-specific applications where accuracy and citations are critical, such as legal research, medical diagnosis, customer support, and enterprise knowledge bases. The architecture consists of two main components: a retrieval component that finds relevant documents from a knowledge base, and a generation component that uses retrieved information to produce coherent, informed responses.